

ALMA GERALDINE HETHARIE

Ambon, Indonesia | +62857-5785-8852 | alma281104@gmail.com |

LinkedIn: www.linkedin.com/in/alma-hetharie462096262 | GitHub: <https://github.com/AlmaGeraldine24> | Portfolio: <https://almageraldine24.github.io/>

SUMMARY

Industrial Engineering student with a strong interest in Artificial Intelligence, Machine Learning, and Data Analytics. Experienced in Python programming, data analysis, and machine learning model development through national certified training programs and project-based learning, including a clustering and classification project on banking transaction data for anomaly detection. Proven ability to work in collaborative environments, conduct data-driven analysis, and contribute to research activities. Passionate about leveraging AI and data-driven solutions to support decision-making, including in the finance and banking sector.

EDUCATION

Universitas Pattimura | 2023 - Present

Bachelor of Industrial Engineering, GPA: 3.73/4.00

Relevant Interests: Artificial Intelligence, Machine Learning, Data Analytics, Data Science

EXPERIENCE

Student Research Collaborator | Universitas Pattimura | 2026 - Present

- Assist lecturers in research activities including data collection, processing, and analysis.
- Support research documentation and report preparation.
- Participate in survey preparation and field data collection activities.
- Collaborate with research teams in organizing and managing research datasets.

PROJECTS

Clustering and Classification Analysis of Banking Transactions for Anomaly Detection | Final Submission Project - Machine Learning for Beginners (BMLP), Dicoding | 2025-2026

- Performed clustering on 2,512 bank customer transaction records using an unsupervised learning algorithm (K-Means) optimized with Principal Component Analysis (PCA) to identify transaction patterns and potential anomalies (fraud detection).
- Built classification models (Decision Tree and Random Forest) based on clustering results to predict transaction categories/labels, including hyperparameter tuning to improve model accuracy.
- Conducted data understanding, data cleaning, and feature engineering (including one-hot encoding), and evaluated models using classification metrics (accuracy, precision, recall, F1-score).
- Implemented the complete end-to-end machine learning project lifecycle, from data exploration to model persistence (.h5) for deployment purposes.
- Technologies: Python, Scikit-learn, Pandas, NumPy, K-Means Clustering, PCA, Decision Tree, Random Forest, Google Colab

HealMate AI - AI-Powered Emotional Recovery Platform | DBS Foundation Coding Camp Capstone Project | 2026

- Collaborated with a multidisciplinary team to develop an AI-powered platform that supports emotional recovery after heartbreak.
- Contributed to project planning, feature development, testing, and documentation.
- Participated in designing AI-based solutions focused on user well-being and engagement.
- Utilized GitHub for collaboration, version control, and project management.
- Presented project outcomes and recommendations to mentors and evaluators.
- Technologies: Python, Machine Learning, GitHub, AI Tools

YouTube Comment Sentiment Analysis | Personal NLP Project | 2026

- Collected and processed YouTube comments through data scraping techniques.
- Applied Natural Language Processing (NLP) methods for text preprocessing and sentiment analysis.
- Performed data cleaning, feature extraction, and analytical interpretation.

- Generated insights from user opinions and engagement patterns.
- Technologies: Python, NLP, Pandas, Scikit-learn

Data Analysis Project | *Asah Program by Dicoding* | 2025

- Conducted data cleaning and preprocessing.
- Performed exploratory data analysis (EDA) to identify trends and patterns.
- Created visualizations and generated insights from datasets.
- Applied analytical methods to support data-driven decision making.
- Technologies: Python, Pandas, NumPy, Matplotlib

TECHNICAL SKILLS

Programming: Python

Machine Learning & AI: Machine Learning Fundamentals, Natural Language Processing (NLP), Model Evaluation, Clustering (K-Means), Classification (Decision Tree, Random Forest)

Data Analytics: Data Cleaning, Data Processing, Exploratory Data Analysis (EDA), Data Visualization

Tools: GitHub, Google Colab, Jupyter Notebook, Microsoft Excel

Soft Skills: Problem Solving, Teamwork, Communication, Critical Thinking, Research, Time Management

CERTIFICATIONS

- Asah Program by Dicoding - Machine Learning Cohort (2025-2026)
- Kominfo - Data Analyst Training (2024)
- Kominfo - Python Programming Training (2023)
- Membangun Proyek Deep Learning Tingkat Mahir - Dicoding (2026)
- Belajar Fundamental Deep Learning - Dicoding (2026)
- Membangun Sistem Machine Learning - Dicoding (2025)
- Bootcamp Data Analytics - Digital Talent Scholarship (2023)
- English for Career Development (Coursera) - University of Pennsylvania (2023)